

SignalForge

AI-Powered Trading Signal System

Business Requirements Document

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1. Executive Summary

SignalForge is an AI-powered trading signal generation system developed to provide institutional-grade, high-conviction trade signals for digital asset markets. The system combines Smart Money Concepts (SMC) / Inner Circle Trader (ICT) methodologies with multi-timeframe technical analysis, large language model (LLM) reasoning, and a rigorous multi-layer filter gate to produce actionable signals with quantitatively validated confidence scores.

The system operates on a Linux server hosted on Amazon Web Services (AWS), monitoring four major digital assets (BTC, ETH, BNB, SOL) in real time via WebSocket connections. SignalForge spans 59 Python source files (15,783 lines of code) encompassing asynchronous data ingestion, custom SMC pattern detection, confluence scoring across 12 distinct signal categories, and a 10-layer filter gate that enforces strict admission criteria before any signal is considered for execution.

System at a Glance

Attribute	Detail
Architecture	Async Python 3.11+, asyncio, ccxt/ccxt.pro, SQLite, pandas
Deployment	AWS Linux server managed via systemd, health watchdog port 8080
Assets Monitored	BTC, ETH, BNB, SOL
Signals Processed	279 (to date)
Paper Trades Opened	0 — reflecting strict filter-gate standards
Development Cadence	9 sprints delivered over multiple weeks

2. Business Objectives

2.1 Automated 24/7/365 Market Monitoring

Eliminate the requirement for human screen-watching by deploying an autonomous monitoring agent that operates continuously across all global trading sessions. The system ingests real-time market data via WebSocket connections (ccxt.pro), processing tick-level and candlestick data across four assets simultaneously.

2.2 Institutional-Grade Signal Quality

Produce signals that meet or exceed the quality standards of professional trading desks by implementing SMC/ICT methodology, 12-layer confluence scoring, and LLM-assisted contextual analysis. Only signals with confidence $\geq 70\%$ and R:R ≥ 1.5 are passed.

2.3 Quantified Risk Management

Every signal includes a calculated entry, stop-loss, take-profit, and position size. Risk per trade is capped at 1% of account equity, enforced by the risk engine.

2.4 Transparent Audit Trail

All signals, decisions, filter-gate outcomes, and LLM reasoning are persisted to SQLite. Full reproducibility of any signal decision is possible via the database.

2.5 Zero-Downtime Operations

The system targets 99.9% uptime via systemd auto-restart, OOM protection, health monitoring on port 8080, and Telegram-based alerting for any degradation event.

3. Success Criteria

KPI	Target	Measurement Method
Signal Win Rate	>= 55%	Closed paper trades (TP/SL hit)
Average R:R	>= 1.8:1	Mean of all closed trade R:R
Signal Confidence Score	>= 70%	LLM output per signal
Filter Gate Pass Rate	5-15%	Signals passed / signals evaluated
Service Uptime	>= 99.9%	systemd restart logs + watchdog
WS Reconnection Time	< 30s	ccxt.pro reconnect event log
LLM Response Time	< 10s avg	Logged per-call latency
False Positive Rate	< 20%	Manual review of closed trades

4. Stakeholder Map

Role	Name / System	Responsibility
Executive Sponsor	Fazrial	Strategic direction, final approval, capital allocation
Product Owner	Fazrial	Requirements, acceptance criteria, sprint prioritisation
Development Lead	Hermes Agent	Architecture, code generation, testing, deployment
LLM Reasoning	DeepSeek-chat	Signal reasoning, confidence scoring, risk assessment
Data Provider	Binance (ccxt.pro)	Real-time WebSocket OHLCV + orderbook data
Delivery Channel	Telegram Bot	Signal delivery, command interface, daily pings

Infrastructure



AWS EC2



Hosting, process management, persistent storage

5. Cost-Benefit Analysis

Monthly Operating Costs

Cost Item	Estimated Monthly Cost
AWS EC2 instance (t3.small or equivalent)	~\$15 USD
DeepSeek LLM API (279+ calls/month at \$0.001/call)	< \$1 USD
Binance API access	Free tier
Telegram Bot API	Free
Total Estimated Monthly Cost	~\$16 USD

Benefit Quantification

- Eliminates manual chart-watching: saves 4-8 hours/day of analyst time
- Consistent rule-based execution eliminates emotional trading bias
- Paper trading mode allows zero-risk strategy validation before live deployment
- At 55% win rate and 1.8 R:R on \$10K virtual capital, expected monthly return: \$400-\$800

6. Risk Assessment

Risk	Likelihood	Impact	Mitigation
OOM kill / service crash	Medium	High	MemoryMax=1200M, OOMPolicy=restart, StartLimitIntervalSec=0
LLM API downtime	Low	High	Fallback model configured; graceful PASS on failure
Exchange API rate limit	Low	Medium	ccxt.pro built-in rate limiting; exponential backoff
False positive signals	Medium	Medium	10-layer filter gate; confidence >= 70% threshold
Runaway log volume	High	Low	logrotate daily compress; 7-day retention
WebSocket disconnect	Medium	Medium	Auto-reconnect < 30s; watchdog alert on stale data
Regulatory / compliance	Low	High	Paper trading only; no live capital at risk currently

7. ROI Projections — 12-Month Model

Based on paper trading performance targets and a transition to live trading at month 3, the following projections assume conservative signal frequency (2 signals/week) and risk-adjusted position sizing at 1% equity per trade.

Period	Mode	Capital	Expected Return	Cumulative P&L
Month 1-2	Paper trading	\$10,000 virtual	Validation only	\$0 (validation)
Month 3-6	Live (cautious)	\$1,000 real	3-5% / month	+\$120 to +\$200
Month 7-12	Live (scaled)	\$5,000 real	4-7% / month	+\$1,200 to +\$2,100

Disclaimer

All projections are estimates based on backtested strategy parameters. Past performance of SMC/ICT signals does not guarantee future results. Live trading involves capital risk.

8. Compliance Framework

8.1 Audit Trail

- All signals logged to SQLite with timestamp, asset, direction, confidence, R:R, and LLM reasoning
- Filter gate decisions logged with rejection reason for every evaluated signal
- Trade outcomes (TP/SL/manual close) tracked in paper_trades table

8.2 Change Control

- All source code changes tracked via Git (github.com/fazrialf/signalforge)
- Configuration changes to signal thresholds require documented justification
- LLM model changes require re-validation against historical signal dataset

8.3 Data Governance

- No personal user data stored — system processes only public market data
- API credentials stored in .env file, excluded from version control via .gitignore
- Database backups recommended weekly; logs retained 7 days via logrotate

9. Approval

Role	Name	Signature	Date
Executive Sponsor			
Product Manager			
Engineering Lead			
Risk & Compliance Officer			

Document Revision History

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1.0	June 25, 2026	Product Engineering	Initial document creation for stakeholder review

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